Jenkins is an open-source automation server that helps automate the parts of software development related to building, testing, and deploying, facilitating continuous integration and continuous delivery (CI/CD). Below is a comprehensive list of common Jenkins parameters along with their definitions and descriptions.

General Jenkins Parameters

Jenkins Job Parameters

- **`String Parameter`**
  - **Definition**: Accepts a string value.
  - **Description**: Useful for specifying file paths, commands, or any text input.

- **`Boolean Parameter`**
  - **Definition**: Accepts a boolean value (true or false).
  - **Description**: Useful for enabling or disabling features.

- **`Choice Parameter`**
  - **Definition**: Allows the user to select a value from a predefined list of choices.
  - **Description**: Useful for selecting from multiple options.

- **`File Parameter`**
  - **Definition**: Allows the user to upload a file.
  - **Description**: Useful for passing files into the job workspace.

- **`Password Parameter`**
  - **Definition**: Accepts a password value.
  - **Description**: Hides the input value and is useful for sensitive information.

- **`Run Parameter`**
  - **Definition**: Allows selection of a specific run of a specified project.

  - **Description**: Useful for triggering builds based on the results of previous runs.

- **`Text Parameter`**
  - **Definition**: Accepts a multi-line text value.
  - **Description**: Useful for providing long-form text inputs, such as scripts.

Jenkins Pipeline Parameters

- **`parameters`**
  - **Definition**: Defines input parameters for a pipeline.
  - **Description**: Specifies the parameters that can be passed to the pipeline.

- **`string`**
  - **Definition**: Defines a string parameter.
  - **Description**: Similar to the String Parameter in Freestyle jobs.

- **`booleanParam`**
  - **Definition**: Defines a boolean parameter.
  - **Description**: Similar to the Boolean Parameter in Freestyle jobs.

- **`choice`**
  - **Definition**: Defines a choice parameter.
  - **Description**: Similar to the Choice Parameter in Freestyle jobs.

- **`file`**
  - **Definition**: Defines a file parameter.
  - **Description**: Similar to the File Parameter in Freestyle jobs.

- **`password`**
  - **Definition**: Defines a password parameter.

  - **Description**: Similar to the Password Parameter in Freestyle jobs.

- **`text`**
  - **Definition**: Defines a text parameter.
  - **Description**: Similar to the Text Parameter in Freestyle jobs.

Jenkins Configuration Parameters

General Configuration

- **`System Message`**
  - **Definition**: A message that appears on the Jenkins dashboard.
  - **Description**: Useful for conveying information to all users.

- **`# of executors`**
  - **Definition**: Number of concurrent builds that Jenkins can perform.
  - **Description**: Defines the number of jobs that can run in parallel.

- **`Home directory`**
  - **Definition**: The directory where Jenkins stores its data.
  - **Description**: Specifies the location of the Jenkins home directory.

- **`Environment variables`**
  - **Definition**: Variables that are available globally within Jenkins.
  - **Description**: Useful for setting global properties that can be accessed in all jobs.

Security Configuration

- **`Enable security`**
  - **Definition**: Turns on security features.

  - **Description**: Enables user authentication and authorization.

- **`Security realm`**

  - **Definition**: Defines how Jenkins users are authenticated.

  - **Description**: Specifies the authentication method (e.g., Jenkins own database, LDAP).

- **`Authorization`**

  - **Definition**: Defines how permissions are granted to users and groups.

  - **Description**: Specifies the access control method (e.g., Matrix-based security, Role-based strategy).

SCM Configuration

- **`Git`**

  - **Definition**: Source Code Management tool integration.

  - **Description**: Configures Jenkins to pull code from a Git repository.

- **`Subversion`**

  - **Definition**: Source Code Management tool integration.

  - **Description**: Configures Jenkins to pull code from a Subversion repository.

Jenkins Build Parameters

Build Triggers

- **`Build periodically`**

  - **Definition**: Schedules builds to run at regular intervals.

  - **Description**: Uses a cron-like syntax to define the schedule.

- **`Build after other projects are built`**

  - **Definition**: Triggers a build after other projects have completed.

  - **Description**: Useful for setting up build dependencies.

- **`GitHub hook trigger for GITScm polling`**
  - **Definition**: Triggers a build when changes are pushed to a GitHub repository.
  - **Description**: Configures a webhook to listen for push events.

- **`Poll SCM`**
  - **Definition**: Periodically checks the SCM for changes.
  - **Description**: Uses a cron-like syntax to define the polling interval.

Jenkins Pipeline Syntax

`pipeline` Block

- **`agent`**
  - **Definition**: Specifies where the pipeline or stage will run.
  - **Description**: Can define specific nodes, Docker containers, or labels.

- **`stages`**
  - **Definition**: Defines a sequence of one or more stages to be executed.
  - **Description**: Encapsulates the different parts of the build process.

- **`stage`**
  - **Definition**: Defines a single stage in the pipeline.
  - **Description**: Each stage can contain multiple steps.

- **`steps`**
  - **Definition**: Defines the individual tasks to be performed within a stage.
  - **Description**: Contains the commands or script to execute.

- **`post`**
  - **Definition**: Defines actions to take after the pipeline or stage completes.
  - **Description**: Can handle success, failure, or always conditions.

- **`environment`**
  - **Definition**: Sets environment variables for the pipeline.
  - **Description**: Makes variables available to all steps in the pipeline.

- **`parameters`**
  - **Definition**: Defines input parameters for the pipeline.
  - **Description**: Specifies the parameters that can be passed to the pipeline.

Jenkins Environment Variables

- **`BUILD_ID`**
  - **Definition**: The current build ID.
  - **Description**: Unique identifier for the current build.

- **`BUILD_NUMBER`**
  - **Definition**: The current build number.
  - **Description**: Incremental build number for the current project.

- **`BUILD_URL`**
  - **Definition**: URL of the build.
  - **Description**: Direct link to the current build.

- **`JOB_NAME`**
  - **Definition**: Name of the project.
  - **Description**: Name of the job being executed.

- **`WORKSPACE`**
  - **Definition**: Path to the workspace directory.
  - **Description**: Directory where the job is executed.

- **`JENKINS_HOME`**
  - **Definition**: Path to the Jenkins home directory.
  - **Description**: Directory where Jenkins is installed and stores its data.

This list covers many of the essential Jenkins parameters. Specific plugins and configurations may introduce additional parameters and options. Always refer to the [Jenkins documentation](https://www.jenkins.io/doc/) for the most comprehensive and up-to-date information.